

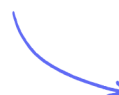
## Multiple Choice Questions

# Organic Chemistry: Alcohols

Alcohols - Introduction / Reactions of Alcohols / Oxidation of Alcohols

|                      |           |
|----------------------|-----------|
| Easy (2 questions)   | /2        |
| Medium (5 questions) | /7        |
| <b>Total Marks</b>   | <b>/9</b> |

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# Easy Questions

1 Which of these compounds is a tertiary alcohol?

- A. 2-methylpropan-2-ol
- B. 3-methylbutan-2-ol
- C. 2,2-dimethylpropan-1-ol
- D. 3,3-dimethylbutan-2-ol

(1 mark)

2 Which reagent would convert an alcohol into an alkene?

- A. acidified potassium dichromate(VI)
- B. anhydrous calcium sulfate
- C. concentrated phosphoric acid
- D. ethanolic potassium hydroxide

(1 mark)

# Medium Questions

1 Which of these compounds does **not** react with acidified potassium dichromate(VI)?

- A.  $\text{CH}_3\text{CH}_2\text{OH}$
- B.  $\text{CH}_3\text{CHOHCH}_3$
- C.  $\text{CH}_3\text{CH}_2\text{CHO}$
- D.  $\text{CH}_3\text{COCH}_3$

(1 mark)

**2 (a)** Organic reactions can be classified in different ways.

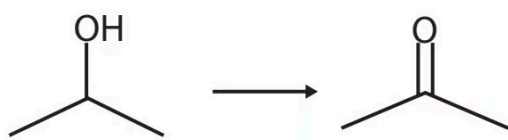
How should the reaction shown be classified?



- A.** addition
- B.** oxidation
- C.** polymerisation
- D.** substitution

**(1 mark)**

**(b)** How should the reaction shown be classified?



- A.** addition
- B.** oxidation
- C.** reduction
- D.** substitution

**(1 mark)**

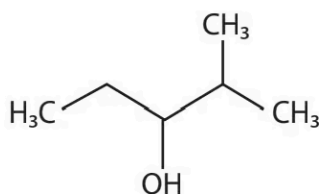
**3 (a)** Some alcohols react with concentrated phosphoric(V) acid,  $\text{H}_3\text{PO}_4$ , to form alkenes.

What type of reaction occurs?

- A.** addition
- B.** elimination
- C.** hydrolysis
- D.** substitution

**(1 mark)**

**(b)** The structure of 2-methylpentan-3-ol is shown.



This alcohol reacts with concentrated phosphoric(V) acid. How many different alkenes can form?

- A.** one
- B.** two
- C.** three
- D.** four

**(1 mark)**

**4** What is the minimum volume of oxygen gas, measured at room temperature and pressure, required for the complete combustion of 9.2 g of  $\text{C}_3\text{H}_8\text{O}_3$  ( $M_r = 92$ )?

[Molar volume of gas at room temperature and pressure =  $24.0 \text{ dm}^3 \text{ mol}^{-1}$ ]

- A.**  $4.8 \text{ dm}^3$
- B.**  $8.4 \text{ dm}^3$
- C.**  $12.0 \text{ dm}^3$
- D.**  $16.8 \text{ dm}^3$

**(1 mark)**

- 5** The oxidation of propan-1-ol by acidified potassium dichromate(VI) forms propanoic acid with a yield of 36% by mass.

What mass of propan-1-ol is needed to form 37.0 g of propanoic acid in this reaction?

[ $M_r$  of propanoic acid = 74     $M_r$  of propan-1-ol = 60]

**A.** 10.8 g

**B.** 36.0 g

**C.** 40.8 g

**D.** 83.3 g

**(1 mark)**